



Material Properties

Large impact in electrical properties of polypropylene by improving the filler-matrix interface effect in PP/PS blends

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ABSTRACT

Inorganic nanoparticles are widely used to improve space charge behavior, DC breakdown strength and other electrical properties of polymer insulating materials, but the uniform distribution of inorganic nanofillers in matrix is difficult due to their agglomeration and bad compatibility with the polymeric matrix. In this paper, polypropylene (PP)/polystyrene (PS) blends were prepared to suppress space charge accumulation and improve DC breakdown strength. Polypropylene-*g*-polystyrene (PP-*g*-PS) graft copolymer was used as compatibilizer to improve the compatibility of PP matrix and PS filler. The evolution of microstructure of PP/PS blends were investigated by scanning electron microscope (SEM), the space charge distributions were measured by a pulsed electro-acoustic (PEA) system, and DC breakdown strength was also tested. The morphologies show that the size of PS particles reduced to 310 nm when the content of PP-*g*-PS graft copolymer increased to 24 wt%, and the interaction between PP matrix and PS particles enhanced. The presence of PS particles in all PP/PS blends suppressed the space charge accumulation compared to neat PP, but the DC breakdown strength in uncompatibilized blend was lower than neat PP. The increasing of content of PP-*g*-PS improved the DC breakdown strength with the maximum value of 408.9 kV/mm was obtained. This may attribute to excellent interface structure formed between PP matrix and PS particles.

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1. Introduction

Space charge accumulation occurred in insulating material when a high direct current (DC) voltage was applied on it, which restricted the development of insulating material, such as high voltage direct current (HVDC) cable [1]. Charges injected into materials were captured by defects, forming charge accumulation, distorting local electric field and accelerating electric breakdown under high electric field [2–4]. Much attention has been paid on suppressing the space charge accumulation [5]. Nanoscale fillers, especially inorganic nanofillers, were widely reported to have great effect on mitigating space charge accumulation [6–11]. Many theories and hypothesis based on interfacial effect have been proposed to explain this phenomenon [12–14]. However, the uniform distribution of

inorganic nanofillers in matrix is difficult due to their agglomeration and bad compatibility with the polymeric matrix. Even surface modification exhibited only limited enhancement on uniform distribution of inorganic nanofillers [15]. Therefore, it is difficult to obtain nanocomposites with desired properties.

Blend is a common method to improve polymer performance, from which homogeneous dispersion is not a problem. Moreover, the immiscible issue between different polymers is easy to solve by adding compatibilizer into blends. Polypropylene (PP) is a commodity polymers widely used for insulative applications, and has been considered as next insulation material for high voltage power cable because XLPE, as a thermoset material, cannot be reused at the end of its lifetime [16–18]. However, space charge accumulation is still a restriction for PP to be used as HVDC plastic cable insulator [19]. Polystyrene (PS) is also an excellent insulation polymer material. Previous study shows no space charge accumulation in PS [20].

In this paper, PP was chosen as matrix and PS was chosen as

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filler to improve space charge distribution and DC breakdown strength. Furthermore, PP/PS blends with tunable interface adhesion were prepared to illustrate the influence of interface on electrical properties. PP and PS are immiscible with serious phase separation in their blends, but their compatibility could be increased by adding compatibilizers [21,22]. So it is easy to obtain a series of PP/PS blends with different interaction between PS particles and PP matrix. In order to avoid introducing other impurities into PP/PS blends, polypropylene-g-polystyrene (PP-g-PS) graft copolymer was selected as compatibilizer. The microstructure of PP/PS blends was studied. DC breakdown strength, space charge distributions, dielectric constant and dielectric loss were discussed.

2. Experimental

2.1. Preparation of PP/PS blends

The polypropylene powder (PP, PPH-XD-045) was produced by Jinjiang Chemical Works, China. Polypropylene-g-polystyrene (PP-g-PS) was synthesized using suspension grafting method in our lab [23]. The composition of PP-g-PS copolymers was shown in Table 1. Polystyrene (PS, 825) was obtained from Liaoning Huajin Chemical Co. Ltd, China. The PP/PS blends were prepared by melt blending with twin-screw extruder. Briefly, PP, PS and PP-g-PS graft copolymer were dried at 80 °C for 24 h before use. Then, they were premixed by a high-speed mixer. The total amount of PS, calculated from both PS homopolymer and PP-g-PS graft copolymer, in all blends is 10 wt%. After that, a SHJ-30 co-rotating twin screw extruder (GIANT MACHINERY, Nanjing, China) was used to prepare PP/PS blends with the temperature from 140 °C to 190 °C. The screws diameter is 30.5 mm. The string extrudate was cooled in water with a water bath and then granulated. The neat PP was also extruded with the same procedure for comparison. The obtained blends were pressed under a pressure of 16 MPa at 190 °C using a flat vulcanizing machine to obtain films for testing. The blends were named as PP-g-PS-x, where x is weight percentage of PP-g-PS copolymers in the blend. For example, PP-g-PS-6 represents that 6 wt% of PP-g-PS copolymers was added into PP/PS blends.

2.2. Scanning electron microscopy

The morphology of PP/PS blends section surface was observed by a scanning electron microscope (SEM, XL 30 ESEM-FEG) operated at 10 kV. The samples were fractured in liquid nitrogen. A thin gold layer was coated on the fracture surfaces at vacuum before observation.

2.3. Particle size analysis

The number-average particle diameter (d_n) of PS particles in blends was evaluated using equation as follow:

$$d_n = \frac{\sum_i n_i d_i}{\sum_i n_i} \quad (1)$$

Herein, d_i represents the diameter of PS particles, and n_i represents the number of PS particles with the diameter of d_i . More than three hundreds of PS particles were analyzed through ImageJ software.

Table 1
The composition of PP-g-PS graft copolymers.

Graft copolymer	PP (wt%)	PS (wt%)	Gr (%)
PP-g-PS	58.8	41.2	7.0

2.4. Differential scanning calorimetry

Thermal properties of the samples were examined using a Perkin-Elmer 7 DSC. Samples were first heated to 180 °C at 10 °C/min and kept for 5 min to eliminate thermal history, then were cooled to 40 °C. The crystallization temperature (T_c) was recorded. The melting temperature (T_m) and melting enthalpy (ΔH_m) were measured during the second heating cycle. The measurements were carried out under a nitrogen atmosphere.

2.5. DC breakdown strength

The DC breakdown strength of samples was characterized with a dielectric strength tester with two plate electrodes at room temperature. The diameter of the upper electrode and the lower electrode are 25 mm and 75 mm, respectively. The thickness of the films was about 75 μm. Samples were immersed in silicone oil to prevent surface flashover. A DC voltage was applied across the electrodes and increased continuously with a rate of 1 kV/s until the films were broken down. Before test, the samples were annealed under 80 °C at vacuum for 12 h to eliminate the impact of the environment and residual stress during sample preparation. The testing ambient temperature is 20 °C, the humidity is 30%. Ten films were tested for each sample. Two-parameter Weibull statistical distribution method was used to treat the breakdown test data. The formula was described as follows [24]:

$$F(E, \alpha, \beta) = 1 - \exp \left[- \left(\frac{E}{\alpha} \right)^\beta \right] \quad (2)$$

Where $F(E, \alpha, \beta)$ is the probability of breakdown, E represents the breakdown strength obtained from experiment, α is the scale parameter representing the characteristic value of dielectric breakdown strength, and β is the shape parameter, reflecting the data scatter.

2.6. Dielectric property test

The dielectric properties of the samples were characterized by a high-performance frequency analyzer (BDS 4000). The samples (100 μm thick) were tested in the frequency range of 1–10⁶ Hz at room temperature. A thin gold layer was evaporated on both sides of the film to serve as electrodes before test. Three samples were tested to determine dielectric properties. The average value was calculated and chosen as final dielectric constant. The representative data of $\tan \delta$ was chosen as final result.

2.7. Space charge distribution

Space charge distribution in PP and PP/PS blends was measured under -40 kV/mm DC electric field for 1800 s at room temperature with a thickness of about 260 μm, using a pulsed electro-acoustic (PEA) system. An electric pulse with an amplitude of 1000 V and a width of 8 ns and output voltage of -20 to 20 kV were provided by the system. The aluminum was evaporated by an evaporation coating machine (ZHD-400) on both sides of the film to serve as electrodes under vacuum. All of the samples were short circuited for 24 h under vacuum at 80 °C. Space charge distributions in PP and PP/PS blends under polarization were recorded.

3. Results and discussion

3.1. Microstructure of PP/PS blends

Aiming at investigating the influence of interface structure on

electric properties, the interface structures between PS particles and PP matrix were changed by adding different content of PP-g-PS graft copolymers. The composition of samples is shown in Table 2. The SEM images of fractured surfaces of PP/PS blends with and without PP-g-PS graft copolymer are presented in Fig. 1. The distributions of PS particle size and the number-average particle

Table 2

PP and PP/PS blends with different content of PP-g-PS graft copolymer content.

Sample	PP(wt%)	PS (wt%)	PP-g-PS (wt%)
PP	100	0	0
PP-g-PS-0	90	10	0
PP-g-PS-6	86.5	7.5	6
PP-g-PS-12	83	5	12
PP-g-PS-18	79.5	2.5	18
PP-g-PS-24	76	0	24

1. The PS in this table represents homogeneous PS, and the content of all PS in blend is 10 wt%.

diameters are shown in Fig. 2. The SEM images show that PS spherically distributes in PP matrix forming sea-island mode, and its size is related to the content of PP-g-PS copolymer. For PP/PS blend uncompatibilized with PP-g-PS copolymer (PP-g-PS-0), the PS particle size was around 500–2000 nm (Fig. 2(a)), causing the number-average diameter of PS nanoparticles up to 950 nm. Meanwhile, there are many cavities and PS particles on the fractured surface. Both cavities and PS nanoparticles appeared at smooth surface, and only a small amount of PS nanoparticles were fractured when quenching the samples at very low temperature (Fig. 1(b)). The situations discussed above deduced bad interfacial adhesion and poor compatibility between PS nanoparticles and PP matrix, which leading to phase separation and weak interphase. The morphology evolutions of PP/PS blends with different content of PP-g-PS copolymer are shown in Fig. 1(c)–(f). When PP-g-PS graft copolymer was added into the blends, the number-average diameters were significantly decreased. The number-average diameter of PS particles in the blend containing 6 wt% of PP-g-PS graft

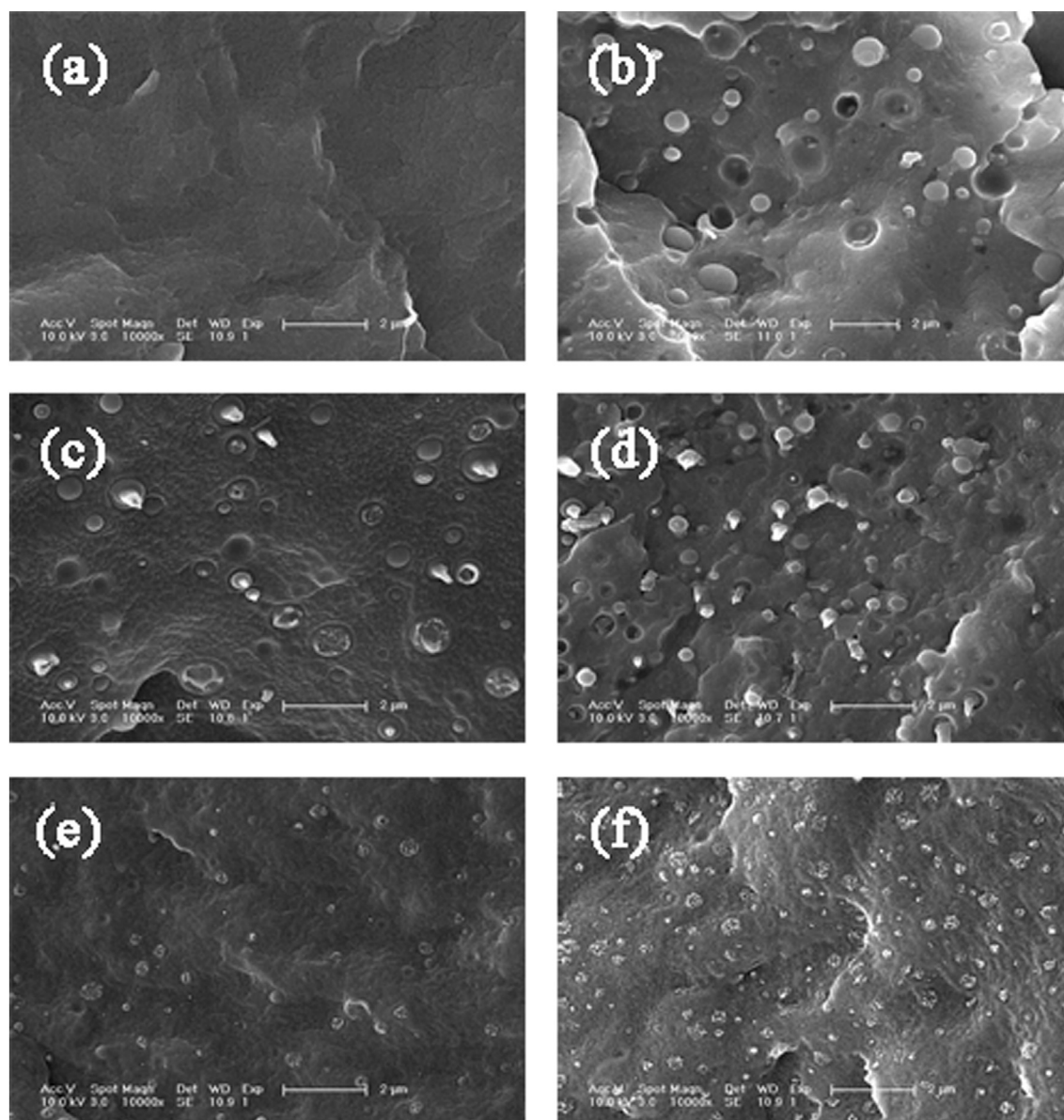


Fig. 1. SEM images of fracture surface of PP and PP/PS blends: (a) PP, (b) PP-g-PS-0, (c) PP-g-PS-6, (d) PP-g-PS-12, (e) PP-g-PS-18, (f) PP-g-PS-24.

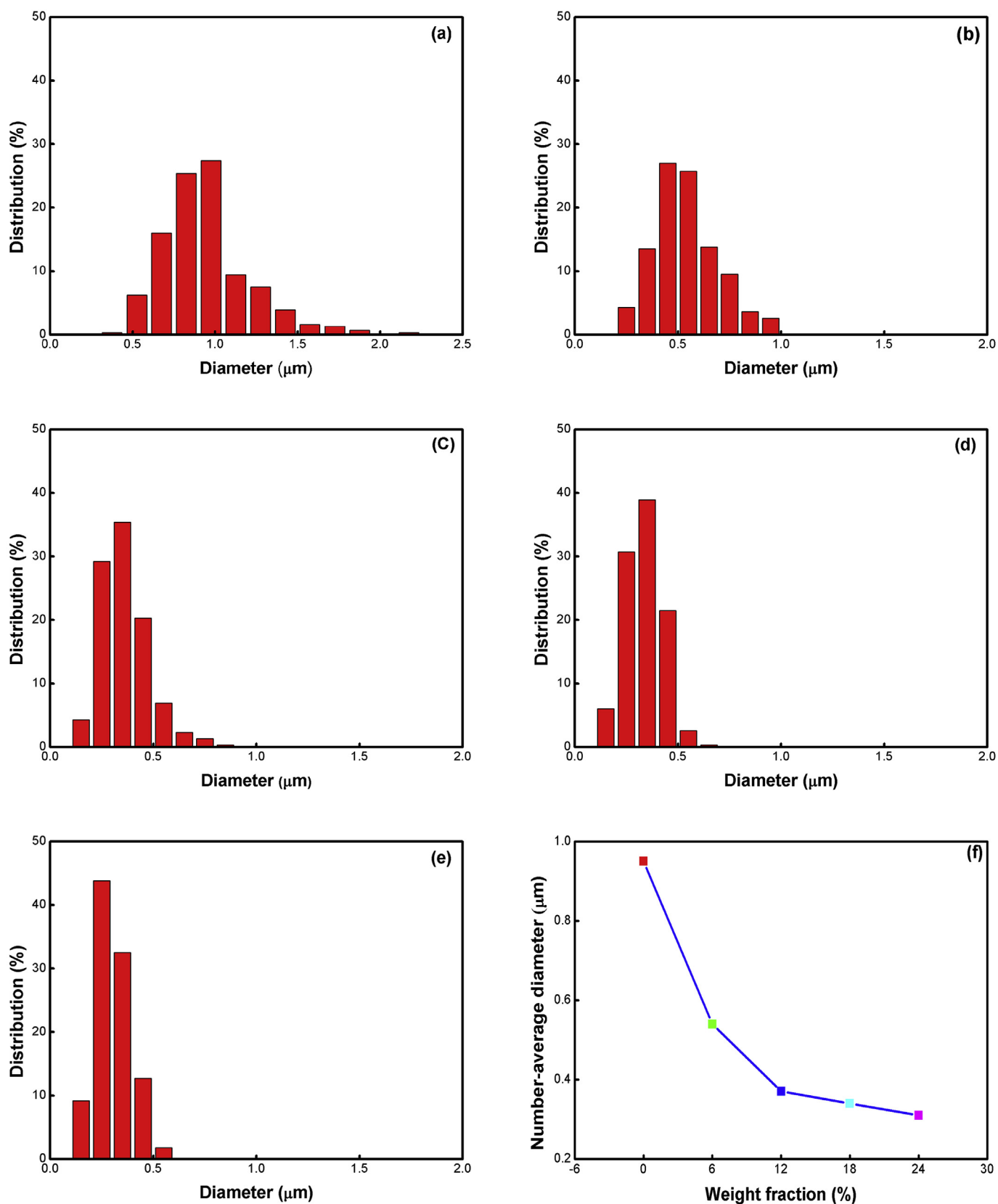


Fig. 2. PS particle diameter distribution and number-average diameter in PP/PS blends with or without PP-g-PS grafting copolymer: (a) PP-g-PS-0, (b) PP-g-PS-6, (c) PP-g-PS-12, (d) PP-g-PS-18, (e) PP-g-PS-24 and (f) number-average diameter of PS particle in PP/PS blends.

copolymer (PP-g-PS-6) was decreased to 540 nm, and the particle size distribution was obviously narrowed (Fig. 2(b)) compared to PP-g-PS-0. With further increasing of content of PP-g-PS copolymer, the number-average diameters of PS particles continue to decrease, and the same with diameter distributions. In addition, compared to PP-g-PS-0, more PS particles in PP-g-PS-6 were fractured, and almost the whole PS particles in PP-g-PS-18 and PP-g-PS-24 were fractured on surface after the samples were quenched in the liquid nitrogen. This means that the interface structure of PP/PS blends were changed by adding PP-g-PS copolymer. The existence of PP-g-PS graft copolymer reduced the interfacial tension, suppressed the phase coalescence and improved the interfacial adhesion, which leading to uniform dispersion of PS phase. Therefore, the size of PS particles was decreased, resulting in larger interface region which may have huge impact on electric properties. The reasons mentioned above were responsible for excellent interphase and the size decreasing of PS particles.

3.2. Thermal properties of PP/PS blends

To investigate the effect of PS and PP-g-PS graft copolymer on PP crystallization process, thermal properties of PP and PP/PS blends were analyzed. The DSC curves of PP and PP/PS blends uncompatibilized and compatibilized with PP-g-PS graft copolymers are shown in Fig. 3(a) and (b), and the important parameters are given in Table 3. As illustrated in Fig. 3 and Table 3, the crystallization temperatures (T_c) of all PP/PS blends were lower than that of neat PP, which indicated that the addition of PS hindered the crystallization of PP. The melting temperatures (T_m) of PP-g-PS-0 was 162.7 °C which was not changed in comparison with neat PP, however, the T_m of compatibilized PP/PS blends were 1–2 °C lower than that of neat PP. To verify the effect of PP-g-PS graft copolymer on crystallinity of PP in compatibilized PP/PS blends, the crystallinity degree (X_c) was calculated and summarized in Table 3. The formula was described as follows [20]:

$$X_c = \frac{\Delta H_m}{\Delta H_{m,PP}^\infty} \times \frac{1}{\Phi(PP)} \times 100\% \quad (3)$$

where ΔH_m is the melting enthalpy of the blends obtained from DSC measurement, $\Delta H_{m,PP}^\infty$ is the melting enthalpy of the 100% crystalline PP ($\Delta H_{m,PP}^\infty = 209$ J/g) referred from previous literature [25], and $\Phi(PP)$ is the weight fraction of PP in the blends. Results show that no obvious difference in X_c value is found between neat PP and PP/PS blend uncompatibilized or compatibilized with

Table 3

DSC and DC breakdown parameters for PP and PP/PS blends.

Sample	T_m (°C)	T_c (°C)	ΔH_m (J/g)	X_c (%)	α (kV/mm)	β
PP	162.5	110.6	99.1	47.4	377.3	10.46
PP-g-PS-0	162.7	109.0	89.7	47.7	325.8	11.98
PP-g-PS-6	161.9	109.6	89.0	47.3	359.5	15.18
PP-g-PS-12	160.2	109.1	89.7	47.4	377.1	18.85
PP-g-PS-18	161.0	109.6	88.4	47.0	391.8	8.04
PP-g-PS-24	161.7	109.1	89.6	47.6	408.9	11.25

PP-g-PS copolymer, which shows same crystallinity of PP in all blends. From the melt crystallization behavior, it was found that the addition of PS can hardly affect PP crystallization, but there existed interactions between PS particles and PP matrix after introducing PP-g-PS graft copolymers into PP/PS blends, since the T_m of compatibilized PP/PS blends was slightly lower than that of neat PP and uncompatibilized PP/PS blends.

3.3. Dielectric properties

The dielectric constant and dielectric loss tangent ($\tan\delta$) of PP and all PP/PS blends are described in Fig. 4. PP-g-PS-0 had highest dielectric constant, and then the dielectric constant decreased by adding PP-g-PS copolymer into blends, with lowest value gained by PP-g-PS-24. This result may be attributed to interface polarization [26]. The interfacial adhesion between PP matrix and PS particles is weaker in PP-g-PS-0 than that in PP/PS blends compatibilized with PP-g-PS copolymer. Therefore, the molecular polarization on interface area became easy in PP-g-PS-0, thus increased dielectric constant. At the same time, the relaxation may occur easily because of the defects and free volume introduced by poor interface, causing an increase in dielectric constant [6,27]. The addition of PP-g-PS copolymer enhances interaction between PP and PS, which limited the probability of dipolar reorientation in electric field, resulting in low dielectric constant of PP/PS blends with PP-g-PS copolymer. This phenomenon was also observed in many inorganic nanocomposites that the dielectric constant of nanocomposites is lower than the base polymer, even the dielectric constant of fillers is much higher than polymer matrix [28–31]. The explain is that the movement of molecular chain is restricted, causing the restriction of material polarization. The dielectric constant was stable in a wide range of frequency. This indicated that the dielectric constant of blends was independent of the variation of frequency. The $\tan\delta$ values of PP/PS blends were also

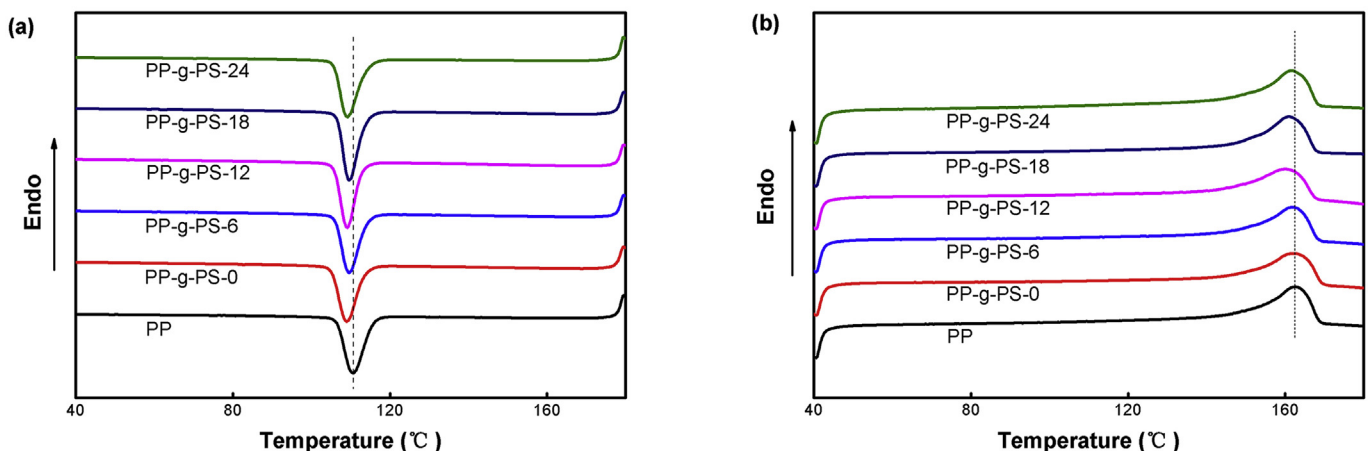


Fig. 3. DSC cooling (a) and heating (b) curves of PP and PP/PS blends.

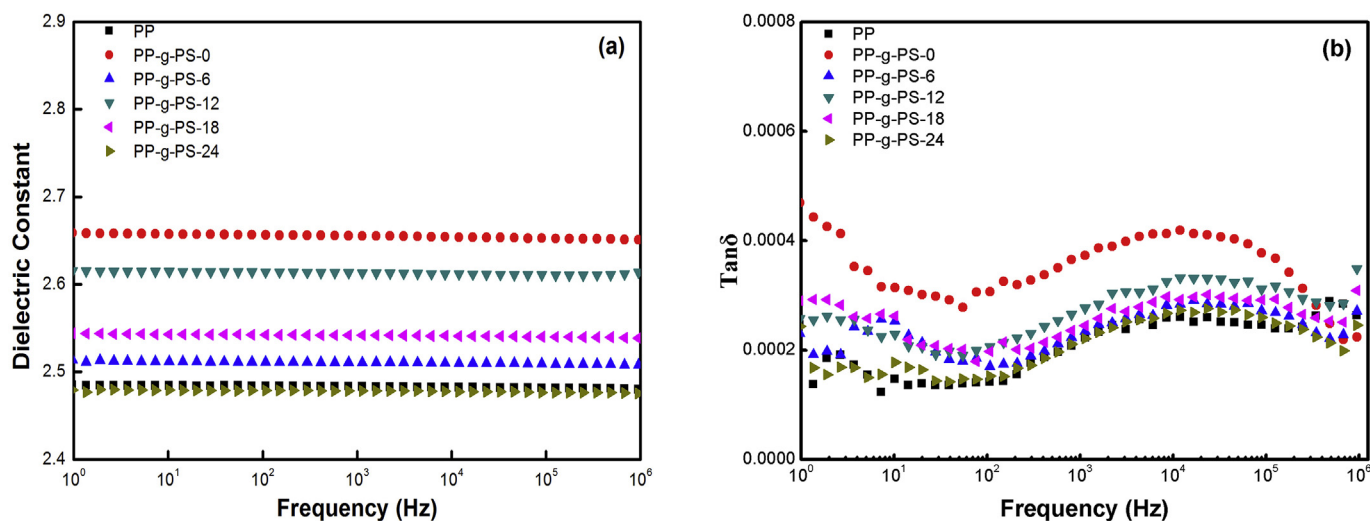


Fig. 4. (a) Dielectric constant and (b) $\tan\delta$ of PP and PP/PS blends.

reduced by compatibilizing with PP-g-PS graft copolymers. From the existing results, it can be concluded that the excellent interface structure will improve dielectric properties, which has advantage for the blends using as insulating material. However, the effect of PP-g-PS content on dielectric behavior is random, which could be found from both dielectric constant and $\tan\delta$. Therefore, it is necessary to study further to find its internal mechanism.

3.4. DC breakdown strength

The Weibull plots of the DC breakdown strength are illustrated in Fig. 5, and the characteristic values of dielectric breakdown strength (α), and the shape parameter (β) are summarized in Table 3. The results indicate that the DC breakdown strength of PP-g-PS-0 decreased obviously, the characteristic value of breakdown strength is 325.8 kV/mm, lower than that of neat PP (377.3 kV/mm). This phenomenon can be explained as follow. PP and PS are thermodynamic immiscible, causing weak interface and poor adhesion between PP matrix and PS particles in PP-g-PS-0 which could be observed clearly from SEM photograph in Fig. 1(b). These conditions would have lead to voids formation at matrix/filler interface

thus trap charge carriers, which then cause a decrease of breakdown strength [32–34]. Park et al. have studied many blend systems about their breakdown strength, they believed that the presence of voids create local electric field distortion which increases the void size and disrupts material structure [35,36].

With the increasing of PP-g-PS graft copolymer in the content range from 6 wt% to 24 wt%, the DC breakdown strength of composites increased continuously. The maximum value 408.9 kV/mm was gained by PP-g-PS-24. Moreover, the values of β for most PP/PS compatibilized blends were also larger than PP, except PP-g-PS-18 that β was lower than PP, meaning the distributions of breakdown strength data were narrowed, representing more uniform properties for the samples.

For many inorganic naondielectrics, a uniform dispersion of fillers and strong interface between polymer and nanoparticles are important to improve the breakdown strength [37,38]. The role is suitable in PP/PS blend. Firstly, the addition of PP-g-PS graft copolymer improved interfacial adhesion and compatibility between PS particles and PP matrix and resulted in a decrease of defects that improved breakdown strength. Secondly, previous study shows that the benzene ring of styrene dissipated energy of electrons via resonance without destroying the structure of insulators [20]. Well interaction is beneficial to charge carriers that migrate from PP matrix into PS particles which may further reduce the electron energy and improve the DC breakdown strength. Thirdly, the destruction of polymer materials in electric field happens easily in amorphous region, because the defects that broke dielectric properties will be expelled out during crystallization progress and located at amorphous region [15,39]. This phenomenon could be obviously observed in electrical tree aging experiments [40]. Thermal properties show that the crystallinity degree is not changed by adding PS and PP-g-PS graft copolymer, meaning PS is located at amorphous region. The appearance of PS particles increased the electrical strength of amorphous region, so the DC breakdown strength increased with the increasing of content of graft copolymers. That is to say, the breakdown strength seriously depends on interfacial condition. Good interface enhances the breakdown strength.

3.5. Space charge behavior

Space charges is produced and accumulated when a DC voltage

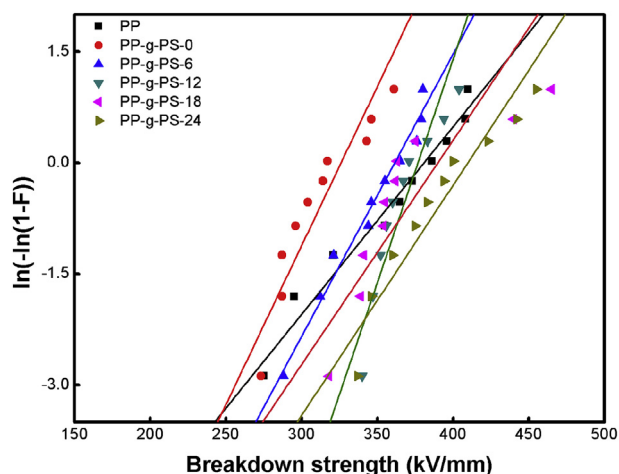


Fig. 5. Weibull plots of DC breakdown strength of PP and PP/PS blends.

is applied on insulators. The accumulation of space charge seriously distorted local electric field, enhanced electrons energy, destroyed insulating materials easily, and ultimately limited the development of high voltage direct current (HVDC) cable. Therefore, space charge suppression is important for insulating materials used under high DC voltage environment.

The space charge distributions during polarization in samples under -40 kV/mm DC electric field for 1800 s at room temperature are shown in Fig. 6. As illustrated from Fig. 6(a), after being polarized under DC electric field for 30 s, heterocharges were found near both cathode and anode in neat PP. After being polarized for 1800 s, the amount of heterocharges near the anode electrode becomes less, while the heterocharges near the cathode electrode transform into homocharges, with a maximum charge density of 3.4 C/m³. The heterocharges observed in insulation materials may attribute to ionization of impurities which were inevitably introduced into composites during manufacturing process. After a high DC field was applied on materials, impurities such as irons or functional groups were ionized or polarized, producing heterocharges which was then accumulated near electrons since the mobility of irons is lower at electric field. With the persistence of polarization, the electrons and holes were injected into samples from both electrodes, which may neutralize the heterocharges initially accumulated near the electrodes. Meanwhile, it could be found that the negative charge injection from cathode is faster than positive charge injection from anode.

The space charge distributions in PP/PS blend are described in Fig. 6 (b)–(f). The space charge accumulation in PP-g-PS-0 was shown in Fig. 6 (b). It is clear that a small amount of heterocharges were observed in the samples near cathode, and almost no space charges were accumulated near anode. After 1800 s, the space charge distribution in PP-g-PS-0 maintained the same, meaning that the space charge was accumulated in a short period of time. With the addition of PP-g-PS copolymer, heterocharges was obviously observed in PP-g-PS-6 near cathode. The density of space charge accumulation is about 3.1 C/m³. The same phenomenon appeared in PP-g-PS-12, furthermore, about 1.5 C/m³ negative space charge was observed in the middle of sample. However, with the further increase of PP-g-PS graft copolymer, the space charge accumulation was reduced in PP-g-PS-18 and PP-g-PS-24.

Space charge comes from charge injection from electrodes or from impurities that polarized under high electric field. Because of the existence of defects such as voids, free volume and dipoles in blend, carrier traps were formed in material. When the electrons or holes move to another electrode under high electric field, they may be trapped by charge traps. Since the depth of traps is different, the charges trapped in shallow traps may de-trap for thermal movement or electrical field force. However, it is difficult for charges that de-trap from deep traps again. When a large amount of charges is trapped, the space charge accumulation occurs in materials.

As illustrated from SEM images (Fig. 1), the average diameter of PS particles in PP-g-PS-0 is as large as 950 nm, and no interaction exists between PP matrix and PS particles. This structure produces free volume and more defects on interface, which may provide carriers a long free path to transmit, so the mobility of the charge carriers was improved [18]. At the same time, interface region introduced by PS particles form more carrier traps. The depth of traps depends on the size of defects, and was deeper as the size of defects becomes smaller [41]. The existence of free volume increased the size of defects, so more shallow traps formed in the interface region, promoting the movement of carriers. This is in consistent with previous study that charge mobility was enhanced by introducing shallow trapping states [9]. Besides, different from

inorganic fillers, charge carriers could be injected into PS particles to form a new path to transmit carriers. Therefore, the resonance structure brought by PS chain could play an important role in charge dispersion that suppressing space charges accumulation. For these reasons, the space charge accumulation in PP-g-PS-0 reduces as show in Fig. 6 (b). The addition of PP-g-PS graft copolymer enhanced the interaction between PS particles and PP matrix, where free volume decreased, thus the size of defects became small which increased the depth of traps. Besides, adding PP-g-PS graft copolymer into blends decrease the particle size, causing the increase of interface region, hence the trap density is also increased. However, the content of PP-g-PS in PP-g-PS-6 and PP-g-PS-12 is small, which resulted in part improvement of interface structure, and this could be observed from SEM images (Fig. 1). That is to say, a good interface structure may be formed in one local space, but a bad interface structure may be formed in another local space. This uneven distribution of the structure causes inhomogeneous trap distribution. So the charges were accumulated easily at the place with large amount of deep traps and space charge accumulation was emerged in PP-g-PS-6 and PP-g-PS-12. With the increase of PP-g-PS graft copolymer, the interface structure was further optimized in PP-g-PS-18 and PP-g-PS-24, and the size of PS particles was further decreases, leading to an increase in the density of PS particles in blends, and hence the distance between PS particles was then reduced. These microstructure changes resulted in a uniform distribution of charge traps, causing a suppression of space charge accumulation. Then the effect of space charge suppression introduced by PS particles may cover whole material that may result in uniform distribution of charges. Meanwhile, the deep carrier traps and the increased trap density caused by the graft copolymer increasing will also restrict the mobility of carrier, which is helpful for space charge suppression in PP-g-PS-18 and PP-g-PS-24.

Three main interfaces exist in polymer blends, including matrix and filler, lamella and amorphous regions and boundaries of spherulites. And many reports show that the boundaries of spherulites are thought to be the key to suppress space charge accumulation [42–44]. However, from this study, it can be clearly found that the interface between matrix and filler also exhibits an important role in improving space charge distribution. And the reason why previous studies did not find the influence of interface between polymer domains on suppressing space charge accumulation may be attributed to the size of blended polymer which is too large that cannot form enough interfaces to influence the distribution of space charge. Moreover, from the results of space charge behavior in this study, we speculate that two mechanisms can suppress space charge accumulation. One is to improve the mobility of carrier to uniform space charge distribution. The other is to reduce carrier mobility to form independent electric field near electrode to prevent carrier injection.

4. Conclusions

In conclusion, the existence of PP-g-PS enhanced interfacial adhesion between PP matrix and PS particles, decreased the size of PS particles, making PS particles uniformly dispersed in PP matrix with nanoscale. The minimum value of 310 nm was observed in PP-g-PS-24. Crystallization behavior of PP didn't change with introducing of PS and PP-g-PS graft copolymer. All Samples have a stable dielectric constant and low dielectric loss tangent in a broad range of frequency. The breakdown strength improved obviously in PP/PS blends through adding PP-g-PS graft copolymer. In all samples, PP-g-PS-24 shows highest DC breakdown strength of 408.9 kV/mm. The excellent interface structure of PP-g-PS-18 and PP-g-PS-24 brings almost no space charge accumulation in both samples. As

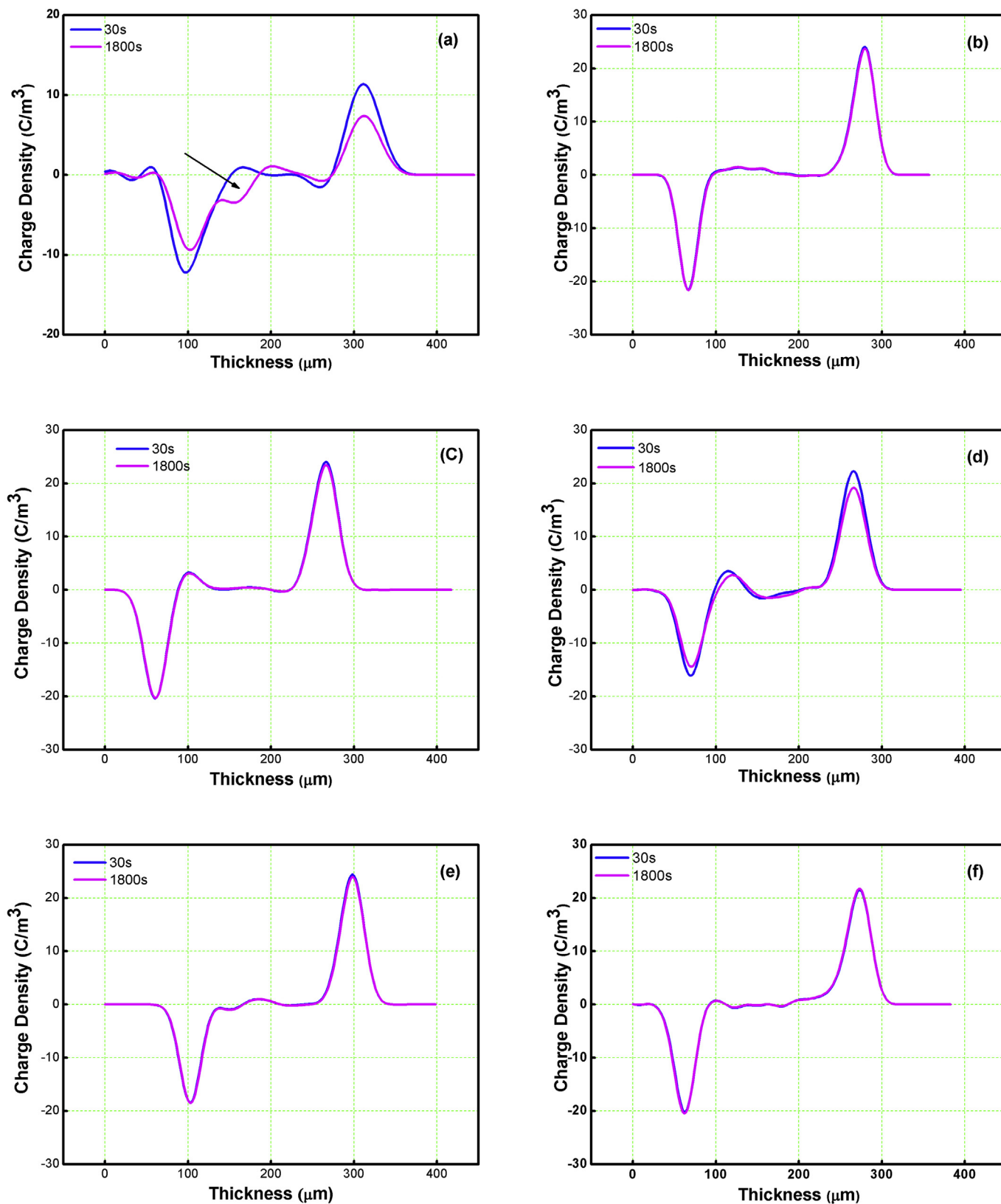


Fig. 6. The space charge distribution in PP and PP/PS blends under -40 kV/mm DC electric field for 30s and 1800s: (a) PP, (b) PP-g-PS-0, (c) PP-g-PS-6, (d) PP-g-PS-12, (e) PP-g-PS-18, (f) PP-g-PS-24.

a result, this research has its potentials for developing advanced thermoplastic insulating materials.

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